

Stack : utility data structure:

LIFO

01 May 2026 20:16

Int arr[5];

[4]	900	116
[3]	700	112
[2]	750	108
[1]	220	104
[0]	11	100
top		
[-1]		

3 basic operations of stack

- 1) Push -> add the element on the stack
- 2) Pop -> delete the element from the stack
- 3) Peek -> check the current element

Push :

- 1) Increment the top by 1
- 2) Insert the element at top position

Pop :

- 1) Decrement the top by 1

Peek :

- 1) Return the value at top position

Stack empty condition :

top == -1

Stack full condition :

top == SIZE-1

push(val)

{
① Increment the top var.
top++;

② Insert the value at the top index.
arr[top] = val;

}

pop()

{
Optional
arr[top] = 0;

top--;

}

int peek()
{
return arr[top];
}